National University of Singapore
School of Computing
CS3244: Machine Learning
Solution to Tutorial 3

Linear Classifiers and Logistic Regression

1. **Logistic Regression**.

   Which of the following evaluation metrics cannot be applied in case of logistic regression output to compare with target? Explain your answer.

   (a) Accuracy
   (b) AUC-ROC
   (c) Log loss
   (d) Mean-Squared-Error

   Mean-Squared Error.

   We use the input $\mathbf{x}$ *to arrive at a continuous prediction* $\hat{y}$ *where for* $\hat{y} > 0$*, we classify that observation as class 1, otherwise to class -1.*
   How do we judge the performance? Ideally, we want to maximize the accuracy, by giving a zero loss if sign of $\hat{y}$ *agree with the sign of the true class, i.e.* $y\hat{y} > 0$*. Otherwise, if we make the wrong prediction, the loss is 1. Let* $z = y\hat{y}$*.*
   The accuracy loss for one sample is $L(z) = ((sign(z) + 1)/2$ *. Optimizing this loss function is hard because it is non-convex and discontinuous. That is why we use other surrogate losses that closely approximate this function.*
   Accuracy and AUC-ROC is generally used for assessing the performance for binary classification algorithm.
   The log loss for one sample is $L(z) = log(exp(-x) + 1)/log(2)$*. As you can see, this loss is convex, and is close enough to the accuracy loss.*
   The square error for one sample is $(y - \hat{y})^2 = (yy - y\hat{y})^2 = (1 - z)^2$ *This is further away from the true accuracy loss.*

2. **Logistic Regression**.

   The truth tables for AND, OR, NOR, XOR gates are shown below, which of them CANNOT be studied using logistic regression?

   AND gate

x_1	x_2	y
1	1	1
1	-1	-1
-1	1	-1
-1	-1	-1

   OR gate

x_1	x_2	y
1	1	1
1	-1	1
-1	1	1
-1	-1	-1

   NOR gate

x_1	x_2	y
1	1	-1
1	-1	-1
-1	1	-1
-1	-1	1

   XOR gate

x_1	x_2	y
1	1	-1
1	-1	1
-1	1	1
-1	-1	-1

   XOR gate. Logistic regression only forms linear decision surface. In AND, OR, NOR gates, positive and negative examples are linearly separable, but in XOR gate, positive and negative

examples are not linearly separable.

3. *LFD*, Exercise 3.4 – **Inducing Noise**.

   [*Hint: You can refer to question 5 for the definition of "the Hat Matrix" H and use some conclusions about from question 5.*]

   Consider a noisy target $y = \theta^{*\top}\mathbf{x} + \epsilon$ for generating the data, where ϵ is a noise term with zero mean and σ^2 variance, independently generated for every samples $(\mathbf{x}, y)$. The expected error of the best possible linear fit to this target is thus σ^2.

   For the data $\mathcal{D} = (\mathbf{x}^{(1)}, y^{(1)}), \cdots, (\mathbf{x}^{(m)}, y^{(m)})$, denote the noise in $y^{(m)}$ as $\epsilon^{(m)}$ and let $\epsilon = [\epsilon^{(1)}, \epsilon^{(2)}, \cdots, \epsilon^{(m)}]^\top$; assume that $\mathbf{X}^\top\mathbf{X}$ is invertible. By following the steps below, show that the expected in-sample error of linear regression with respect to $\mathcal{D}$ is given by:

$$\mathbb{E}[E_{in}(\theta_{lin})] = \sigma^2(1 - \tfrac{n+1}{m})$$

   (a) Show that the in-sample estimate of $\mathbf{y}$ is given by $\hat{\mathbf{y}} = \mathbf{X}\theta^* + \mathbf{H}\epsilon$.

   Let's first make clear several concepts and the purpose of this question.

   *(1) What is **in-sample estimate**? In-sample estimate means that suppose we already got a trained model $y = \theta^\top\mathbf{x}$, what is the output when we input all the **training data** into this model. Similarly, **out-sample estimate** means the output of trained model when we input data out of the training set (i.e. data from test set or unseen data).*

   *(2) What is **in-sample error**? The prediction of the trained model is not always equal to the ground truth. So the difference between the in-sample estimate and the ground-truth of the **training data** is the in-sample error. Similarly, out-sample error is the difference between the out-sample estimate and the ground-truth of the **test data**.*

   (3) What is the role of the "noisy target" $y = \theta^{*\top}\mathbf{x} + \epsilon$ in this question? Actually, the whole procedure this question expresses is that: first, we generate a training set by sampling some instances (i.e., $\mathcal{D}$) from this "noisy target" $y = \theta^{*\top}\mathbf{x}+\epsilon$; second, we train a linear regression model ($y = \theta^\top\mathbf{x}$) based on the generated training set (which is noisy). It is obvious that our trained linear regression model cannot perfectly fit the training data, since the training data has additional noise; finally, you're required to calculate the in-sample estimate and in-sample error of the trained model on the generated training set (D).

   Note that in the first line of the description of this problem, the noisy target $y = \theta^{*\top}\mathbf{x}+\epsilon$ refers to one single data instance. This question asks about $\mathbf{y}$, the vector of all of the data instances.

   For a sample instance, we have $y = \theta^{*\top}\mathbf{x} + \epsilon$. By "stacking" all m of the examples on top of each other into a matrix form, we obtain: $\mathbf{y} = \mathbf{X}\theta^* + \epsilon$ (here $\mathbf{y}$ and ϵ are vectors of m outputs and error values, one for each point, respectively). To obtain the in-sample estimate, $\hat{\mathbf{y}}$, we need to input all the training instances into our trained model. The training instances are $\mathbf{X}$, but how to get the trained model? Since we need a linear regression model $\mathbf{y} = \mathbf{X}\theta$, remember that linear regression has a closed-form solution, i.e., Normal Equation (refer to week4 post slides p12-p20) , that $\theta = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y}$. And note that the description of this question mentions that $\mathbf{X}^\top\mathbf{X}$ is invertible, which

guarantees that the closed-form solution is applicable(otherwise we can only use gradient descent to train the model). So the in-sample estimate is:

$$\begin{aligned}
\hat{\mathbf{y}} &= \mathbf{X}\theta && \textit{the linear regression model we trained} \\
&= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{y} && \textit{substitutes } \theta \textit{ with Normal Equation} \\
&= \mathbf{H}\mathbf{y}
\end{aligned}$$

Note that the coefficient of $\mathbf{y}$ *is the hat matrix* $\mathbf{H}$*, which is the multiplication of* $\mathbf{X}$ *and its* ***pseudo inverse***
$(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ *(refer to week4 post slide p19). This also demonstrates how to derivate the hat matrix.*

$$\begin{aligned}
\hat{\mathbf{y}} &= \mathbf{H}(\mathbf{X}\theta^* + \epsilon) && \textit{substitutes } \mathbf{y} \textit{ with } (\mathbf{X}\theta^* + \epsilon) \\
&= \mathbf{H}\mathbf{X}\theta^* + \mathbf{H}\epsilon && \textit{expand} \\
&= \mathbf{X}\underline{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}}\theta^* + \mathbf{H}\epsilon \\
&= \mathbf{X}\theta^* + \mathbf{H}\epsilon && (\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{X}^\top\mathbf{X}) = \mathbf{I}
\end{aligned}$$

<u>Aside.</u> The hat matrix gets its name from this property. It is the matrix that we multiply the original outputs $\mathbf{y}$ *to obtain the predicted values* $\hat{\mathbf{y}}$*; i.e.,* $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$*. We can say* $\mathbf{H}$ *maps a column vector* $\mathbf{y}$ *into the space spanned by* $\mathbf{X}$*.*

This is easier to understand graphically. In Figure 1, the linear regression hyperplane is represented by the plane at the "bottom" defined by $\mathbf{X}$*. Let's look at a particular instance.* $\mathbf{y}$ *is the observed output, here a bit above the plane.* $\mathbf{H}\mathbf{y} = \hat{\mathbf{y}}$ *is thus the point on the linear regression plane that is the projection of* y *onto* $\mathbf{X}$*. It is an* **orthogonal** *projection, as* $\hat{\mathbf{y}}$ *is the point that represents the shortest distance to* $\mathbf{y}$*.*

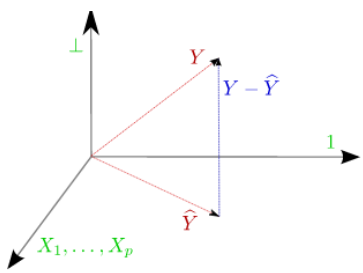


Figure 1. Linear regression as projection. Taken from `http://web.stanford.edu/class/stats191/notebooks/figs/axes_multiple_full.svg`.

(b) Show that the in-sample error vector $\hat{\mathbf{y}} - \mathbf{y}$ can be expressed by a matrix times ϵ. What is the matrix?

From Part (a), we have:

$$\begin{aligned}
\hat{\mathbf{y}} - \mathbf{y} &= \mathbf{X}\theta^* + \mathbf{H}\epsilon - (\mathbf{X}\theta^* + \epsilon) && \textit{Expand} && (1) \\
&= (\mathbf{H} - \mathbf{I})\epsilon. && \textit{Cancel and factor } \square && (2)
\end{aligned}$$

Related, the matrix $\mathbf{M} = (\mathbf{I} - \mathbf{H})$ *(cf Problem 5 (c)) is the orthogonal projection matrix that projects* $\mathbf{y}$ *into its residual* $\hat{\mathbf{y}} - \mathbf{y}$*. This is the vertical line marked in blue, in Figure 1. You can see that* $\mathbf{y} = \mathbf{H}\mathbf{y} + \mathbf{M}\mathbf{y}$*.*

(c) Express $E_{in}(\theta_{lin})$ in terms of ϵ using (b), and simplify the expression using Problem 5 (c).

$E_{in}(\theta_{lin}) = \frac{1}{m}(\hat{\mathbf{y}} - \mathbf{y})^\top(\hat{\mathbf{y}} - \mathbf{y}) = \frac{1}{m}(\epsilon^\top(\mathbf{H} - \mathbf{I})^\top(\mathbf{H} - \mathbf{I})\epsilon)$.

It can be seen that (H-I) is also symmetric, also according to Problem 5 (b). Expanding the above equation and rearranging, we have:

$$E_{in}(\theta_{lin}) = \frac{1}{m}(\epsilon^\top(\mathbf{H} - \mathbf{I})^\top(\mathbf{H} - \mathbf{I})\epsilon) \tag{3}$$

$$= \frac{1}{m}(\epsilon^\top(\mathbf{H} - \mathbf{I})^2\epsilon) \qquad \textit{Using symmetric property}, (\mathbf{H} - \mathbf{I})^\top = (\mathbf{H} - \mathbf{I}) \tag{4}$$

From the above equation 4, if we minimize the $(\mathbf{H} - \mathbf{I})^2$ *term, similar to the proof of Problem 5 (c),*

$$\begin{aligned}
(\mathbf{H} - \mathbf{I})^2 &= (\mathbf{H} - \mathbf{I})(\mathbf{H} - \mathbf{I}) && \textit{Expand} \\
&= \mathbf{H}^2 - 2\mathbf{H}\mathbf{I} + \mathbf{I}^2 && \textit{Multiply out} \\
&= \mathbf{H} - 2\mathbf{H} + \mathbf{I} && \mathbf{I} \textit{ and } \mathbf{H} \textit{ are idempotent}; \textit{Problem 5 (b)} \\
&= \mathbf{I} - \mathbf{H}
\end{aligned}$$

Hence, we can rewrite equation 4 as the following:

$$E_{in}(\theta_{lin}) = \frac{1}{m}(\epsilon^\top(\mathbf{I} - \mathbf{H})\epsilon)$$

(d) Prove that $\mathbb{E}[E_{in}(\theta_{lin})] = \sigma^2(1 - \frac{n+1}{m})$ using (c) and the independence of $\epsilon^{(1)}, \cdots, \epsilon^{(m)}$. [*Hint: The sum of diagonal elements of a matrix (the trace) will play a role. See Problem 5 (d).*]

Continuing from (c), we have

$$E_{in}(\theta_{lin}) = \frac{1}{m}(\epsilon^\top(\mathbf{I} - \mathbf{H})\epsilon) \tag{5}$$

$$= \underbrace{\frac{1}{m}\epsilon^\top\epsilon}_{left} - \underbrace{\frac{1}{m}\epsilon^\top\mathbf{H}\epsilon}_{right} \tag{6}$$

Before proceeding further, remember that the noise(ϵ) *is drawn from Gaussian distribution* $\epsilon \sim \mathcal{N}(0, m\sigma^2)$*. This indicates, the mean,* $\mathbb{E}[\epsilon] = 0$ *and variance,* $var(\epsilon) = m\sigma^2$*.*

Now, let's proceed with the left part of equation 6*:*

$$\begin{aligned}
\mathbb{E}[\frac{1}{m}\epsilon^\top\epsilon] &= \frac{1}{m}\mathbb{E}[\epsilon^2] \\
&= \frac{1}{m}(var(\epsilon) + \mathbb{E}[\epsilon]^2 && Using,\ var(\mathbf{a}) = \mathbb{E}[\mathbf{a}^2] - \mathbb{E}[\mathbf{a}]^2 \\
&= \frac{1}{m}(m\sigma^2 + 0) && var(\epsilon) = m\sigma^2 \ and\ \mathbb{E}[\epsilon] = 0 \\
&= \sigma^2
\end{aligned}$$

and for the right part of equation 6*, we have:*

$$\frac{1}{m}\epsilon^\top\mathbf{H}\epsilon = \frac{1}{m}\{\sum H_{ii}(\epsilon^{(i)})^2 + \sum_{i\neq j} H_{ij}\epsilon^{(i)}\epsilon^{(j)}\} \tag{7}$$

Now for left part of this above equation 7*,*

$$\begin{aligned}
\mathbb{E}[\sum H_{ii}\epsilon_i^2] &= \sum H_{ii}\mathbb{E}[(\epsilon^{(i)})^2] \\
&= trace(\mathbf{H})\ \ \sigma^2 \\
&= (n+1)\sigma^2 && from\ Problem\ 5\ (d),\ trace(\mathbf{H}) = n+1
\end{aligned}$$

We also note since ϵ_i *and* ϵ_j *are independent. So from the right part of the equation* 7*:*

$$\begin{aligned}
\mathbb{E}[\sum_{i\neq j} H_{ij}\epsilon^{(i)}\epsilon^{(j)}] &= \mathbb{E}[\sum H_{ii}]\mathbb{E}[\epsilon^{(i)}]\mathbb{E}[\epsilon^{(j)}] && Using\ \mathbb{E}[\mathbf{ab}] = \mathbb{E}[\mathbf{a}]\mathbb{E}[\mathbf{b}] \\
&= \mathbb{E}[\sum H_{ii}] * 0 * 0 && \mathbb{E}[\epsilon] = 0 \\
&= 0
\end{aligned}$$

We can rewrite equation 7 *as the following :*

$$\begin{aligned}
\frac{1}{m}\epsilon^\top\mathbf{H}\epsilon &= \frac{1}{m}\{\sum H_{ii}(\epsilon^{(i)})^2 + \sum_{i\neq j} H_{ij}\epsilon^{(i)}\epsilon^{(j)}\} \\
&= (n+1)\sigma^2 + 0 \\
&= (n+1)\sigma^2
\end{aligned}$$

Rearranging the above equations, from equation 6 *we get:*

$$\mathbb{E}_D[E_{in}(\theta_{lin})] = \sigma^2(1 - \frac{n+1}{m})$$

For the expected out-of-sample error, we take a special case which is easy to analyze. Consider a test data set $\mathcal{D}_{test} = (\mathbf{x^{(1)}}, y^{(1)'}), \cdots, (\mathbf{x}^{(m)}, y^{(m)'})$, which shares the same input vectors $\mathbf{x}^{(m)}$ with $\mathcal{D}$ but with a different realization of the noise terms. Denote the noise in $y^{n'}$ as $\epsilon^{n'}$ and let $\epsilon^{'} = [\epsilon^{(1)'}, \cdots, \epsilon^{(m)'}]^\top$. Define $E_{test}(\theta_{lin})$ to be the average squared error on $\mathcal{D}_{test}$.

(e) Prove that $\mathbb{E}[E_{test}(\theta_{lin})] = \sigma^2(1 + \frac{n+1}{m})$.

The proof is similarly developed like in (d). We have:

$$\hat{\mathbf{y}} - \mathbf{y} = \mathbf{H}\epsilon - \mathbf{I}\epsilon^{'}$$

Therefore, out of sample error E_{test}:

$$E_{test}(\theta_{lin}) = \frac{1}{m}\|\hat{\mathbf{y}} - \mathbf{y}\| = \frac{1}{m}(\mathbf{H}\epsilon - \mathbf{I}\epsilon^{'})^{\top}(\mathbf{H}\epsilon - \mathbf{I}\epsilon^{'}) = \frac{1}{m}\{\epsilon^{\top}\mathbf{H}\epsilon - \epsilon^{\top}\mathbf{H}\epsilon^{'} - \epsilon^{'T}\mathbf{H}\epsilon + \epsilon^{'T}\epsilon^{'}\}$$

from (d) we have $\mathbb{E}[\epsilon^{\top}\mathbf{H}\epsilon] = \sigma^2(n+1)$ and $\mathbb{E}[\epsilon^{'T}\epsilon^{'}] = m\sigma^2$
Since $\epsilon^{'}$ and ϵ are i.i.d., $\mathbb{E}[\epsilon^{\top}\mathbf{H}\epsilon'] = 0$ and $\mathbb{E}[\epsilon^{'T}\mathbf{H}\epsilon] = 0$. Rearranging the above equations, we prove:

$$\mathbb{E}[E_{test}(\theta_{lin})] = \sigma^2(1 + \frac{n+1}{m})$$

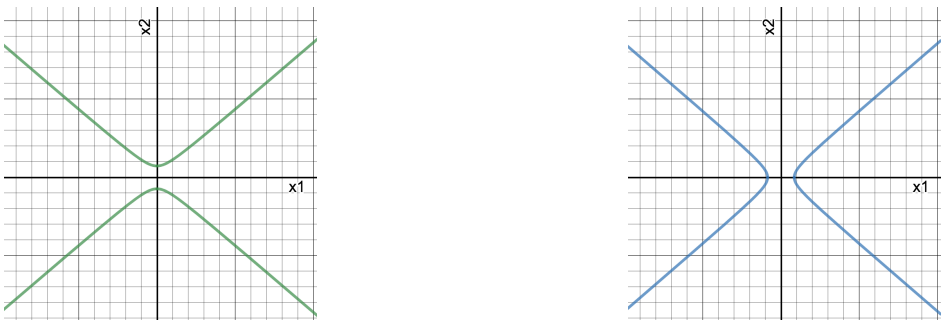
We have thus proved that the in-sample and out-of-sample error of linear regression is bounded to a specific value (given simple assumption of the noise item as a Gaussian).

4. **Nonlinear Transformations.**

Consider the feature transform $\Phi(1, x_1, x_2) = (1, x_1^2, x_2^2)$. Draw and show the boundary (not strictly) in $\mathcal{X}$ that a hyperplane $\hat{\theta}$ in $\mathcal{Z}$ correspond to under the following cases:

check out the webcast for further details

(a) $\hat{\theta}_1 > 0, \hat{\theta}_2 < 0$.
When $\hat{\theta}_0 > 0$, the plot shows a "North-South opening hyperbola"; when $\hat{\theta}_0 < 0$, it's a "East-West opening hyperbola" as shown in Figure 1.



(a) Borders when $\hat{\theta}_0 > 0$ (b) Borders when $\hat{\theta}_0 < 0$

Figure 1: Borders for the two cases of $\hat{\theta}_0$

(b) $\hat{\theta}_1 > 0, \hat{\theta}_2 = 0$.
When $\hat{\theta}_0 > 0$, there's no border in $\mathcal{X}$ space, all data points $\mathbf{x}$ are classified into $y = +1$ category. When $\hat{\theta}_0 < 0$, the border is a single point at x_1 axis as shown in Figure 2.

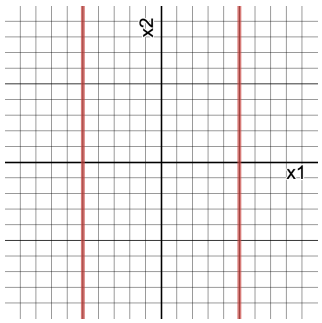


Figure 2: Borders when $\hat{\theta}_0 < 0$

(c) $\hat{\theta}_1 > 0, \hat{\theta}_2 > 0, \hat{\theta}_0 < 0$.
It's either a ellipse or a circle, as shown in Figure 3

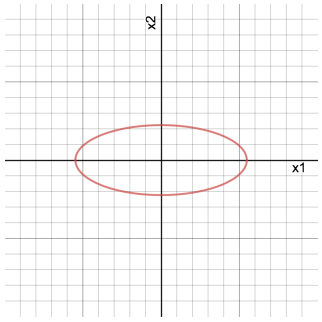


Figure 3: Borders when $\hat{\theta}_0 < 0$

(d) $\hat{\theta}_1 > 0, \hat{\theta}_2 > 0, \hat{\theta}_0 > 0$.
There's no border in $\mathcal{X}$ space. All data points $\mathbf{x}$ are classified into $y = +1$ category as well.

5. *(Optional) LFD*, Exercise 3.3 – **The Hat Matrix.**

The hat matrix is an integral part of understanding linear regression. We see the hat matrix makes an appearance in the analytical, closed form solution of linear regression, as it contains the pseudo inverse $(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$. Let's look at its properties.

Consider the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$, where $\mathbf{X}$ is an m by $n+1$ matrix, and $\mathbf{X}^\top\mathbf{X}$ is invertible.

(a) Show that $\mathbf{H}$ is symmetric.

Show $\mathbf{H}^\top = \mathbf{H}$.

$$\begin{aligned}
\mathbf{H}^\top &= (\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top)^\top && \textit{Expand} && (8)\\
&= (\mathbf{X}^\top)^\top(\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}})^\top && \textit{Use } (\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top && (9)\\
&= \mathbf{X}((\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}})^\top\mathbf{X}^\top && \textit{Use } (\mathbf{AB})^\top = \mathbf{B}^\top\mathbf{A}^\top && (10)\\
&= \mathbf{X}((\mathbf{X}^\top\mathbf{X})^\top)^{-\mathbf{1}})\mathbf{X}^\top && \textit{Use } (\mathbf{A}^\top)^{-\mathbf{1}} = (\mathbf{A}^{-\mathbf{1}})^\top && (11)\\
&= \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top. && && (12)\\
&= \mathbf{H}. && \square && (13)
\end{aligned}$$

(b) Show that $\mathbf{H}^K = \mathbf{H}$, for any positive integer K.

We can prove this by induction. Note that the base case must involve matrix multiplication, so we cannot start from $k = 1$, but should start from $k = 2$.

<u>*Base Case*</u> *($k = 2$): Show* $\mathbf{H^2} = \mathbf{H}$.

$$\begin{aligned}
\mathbf{H^2} &= (\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top)(\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top) && \textit{Expand} && (1)\\
&= (\mathbf{X}\underbrace{(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}}_{A}\underbrace{(\mathbf{X}^\top\mathbf{X})}_{B}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top) && \textit{Associate} && (2)\\
&= (\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top) && \textit{Cancel } A \text{ \& } B \textit{ to } \mathbf{I}, \textit{ drop } \mathbf{I} && (3)\\
&= \mathbf{H}. && \square && (4)
\end{aligned}$$

<u>*Inductive Step*</u>*: Given k is true, show for $k + 1$. Show* $\mathbf{H^{k+1}} = \mathbf{H}$.

$$\begin{aligned}
\mathbf{H^{k+1}} &= (\mathbf{H^k})(\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top) && \textit{Expand} && (5)\\
\mathbf{H^{k+1}} &= (\mathbf{H})(\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top) && \textit{Premise} && (6)\\
&= (\mathbf{X}\underbrace{(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}}_{A}\underbrace{(\mathbf{X}^\top\mathbf{X})}_{B}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top) && \textit{Associate} && (7)\\
&= (\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top) && \textit{Cancel } A \text{ \& } B \textit{ to } \mathbf{I}, \textit{ drop } \mathbf{I} && (8)\\
&= \mathbf{H}. && \square && (9)
\end{aligned}$$

This property $\mathbf{X} = \mathbf{X}^k$ *is called* **idempotency**. *In fact, orthogonal projection matrices (which* $\mathbf{H}$ *is a member of), exactly have both the symmetric and idempotent properties (i.e., "if and only if")*

(c) If $\mathbf{I}$ is the identity matrix of size N, show that $(\mathbf{I}-\mathbf{H})^K = \mathbf{I}-\mathbf{H}$ for any positive integer K.

We can prove this similar to Part (b), using an inductive proof. We just detail the core part here.

$$
\begin{aligned}
(\mathbf{I}-\mathbf{H})^{\mathbf{2}} &= (\mathbf{I}-\mathbf{H})(\mathbf{I}-\mathbf{H}) && \textit{Expand} && (1)\\
&= \mathbf{I^2} - \mathbf{2IH} + \mathbf{H^2} && \textit{Multiply out} && (2)\\
&= \mathbf{I} - \mathbf{2IH} + \mathbf{H} && \mathbf{I}\ \textit{and}\ \mathbf{H}'\textit{s idempotency;}\ \textit{Part}\ (b) && (3)\\
&= \mathbf{I} - \mathbf{H}. && \textit{Drop}\ \mathbf{I}, \textit{associate}\ \square && (4)
\end{aligned}
$$

You might guess that $(\mathbf{I}-\mathbf{H})$ *also is an orthogonal projection matrix, since it is also symmetric and idempotent. And you'd be right.*

(d) (Optional) Show that $trace(\mathbf{H}) = n+1$, where the *trace* (tr) is the sum of diagonal elements. [*Hint:* $trace(\mathbf{AB}) = trace(\mathbf{BA})$.]

$$
\begin{aligned}
trace(\mathbf{H}) &= trace(\underbrace{\mathbf{X}}_{\mathbf{A}}\underbrace{(\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top}_{\mathbf{B}}) && \textit{Expand} && (1)\\
&= trace((\mathbf{X}^\top\mathbf{X})^{-\mathbf{1}}\mathbf{X}^\top\mathbf{X}) && \textit{Apply hint} && (2)\\
&= trace(\mathbf{I}_{(n+1)}). && \underbrace{\mathbf{X}^{-\mathbf{1}}}_{(n+1)\times m}\ \underbrace{\mathbf{X}}_{m\times(n+1)} = \mathbf{I} && (3)
\end{aligned}
$$

For a symmetric and idempotent matrix A, $rank(A) = trace(A)$. *We'll use these properties in the next question.*

6. *(Optional)* **Error bound**.

In classification problems, we have $err_{in} = [sign(\theta^\top\mathbf{x}) \neq y]$ for $y \in \{-1, +1\}$. Which of below is/are the upper bound of err_{in}? Explain your answer.

(a) $err_{in}(\theta) = max(0, 1 - y\theta^\top\mathbf{x})$.

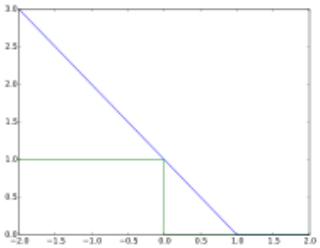
True, it is an upper bound to the true loss 0/1 function.

When $sign(\theta^\top\mathbf{x}) = sign(y)$, $err_{in} = 0$, $err_{in}(\theta) \geq 0$; *when* $sign(\theta^\top\mathbf{x}) \neq sign(y)$, $-y\theta^\top\mathbf{x} > 0$ *and so that* $1 - y\theta^\top\mathbf{x} > 1 = err_{in}$. *This is the* **Hinge Loss** *function, commonly used in the training of support vector machines.*

Hinge Loss (blue) versus 0/1 loss (green). Taken from https://en.wikipedia.org/wiki/Hinge_loss.

(b) $err_{in}(\theta) = max(0, 1 - y\theta^\top\mathbf{x})^2$.

True, as can be seen from (a).



(c) $err_{in}(\theta) = \sigma(-y\theta^\top \mathbf{x})$, where $\sigma(s) = \frac{exp(s)}{1+exp(s)}$.

False. Under a single dimension space, we can see from the plot of sigmoid function that $\sigma(s) < 1$.

(d) $err_{in}(\theta) = max(0, -y\theta^\top \mathbf{x})$.

False. Consider when $sign(\theta^\top \mathbf{x}) \neq sign(y)$ *and uni-dimension situation. One can draw the plot of* $f(\theta) = -y\theta x$*, which is not always greater than 1.*

(e) $err_{in}(\theta) = exp(-y\theta^\top \mathbf{x})$.

True. When $sign(\theta^\top \mathbf{x}) = sign(y)$*,* $err_{in} = 0$*,* $exp(-y\theta^\top \mathbf{x}) > 0$*; when* $sign(\theta^\top \mathbf{x}) \neq sign(y)$*,* $-y\theta^\top \mathbf{x} > 0$*, so that* $exp(-y\theta^\top \mathbf{x}) > 1 = err_{in}$.